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Black Hole Thermodynamics from Counting Convex Subsets of a Lattice

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We derive the Bekenstein-Hawking area law and Hawking temperature from a purely combinatorial computation on discrete lattices, without invoking quantum field theory on curved spacetime. The key input is a *dimension law*: the number of causally convex subsets of the d-dimensional grid $[m]^d$ grows as $\exp(\Theta(m^{d-1}))$, where the exponent $d-1$ detects the spacetime dimension. This law is proved rigorously for all $d \geq 2$ (with formal verification in [16]). Restricting to *cylindrical* (time-symmetric) convex subsets — those of the form $C \times [t_1, t_2]$ where C is a convex subset of the spatial lattice — reduces the exponent by one, yielding entropy scaling as m^{d-2}: the spatial area. This restriction is a kinematic identification of equilibrium configurations, requiring no dynamical input. For $d = 4$ (3+1 spacetime), this recovers the Bekenstein-Hawking relation $S \sim A$. Conversely, the width scaling law implies that any discrete spacetime whose equilibrium entropy obeys the area law must have width $\sim n^{3/4}$, the hallmark of 4-dimensional structure: the dimension is not assumed but forced by the entropy scaling. The Hawking temperature $T \sim 1/m$ follows from one derivative, verified by the computed data. Matching the entropy coefficient to $S = A/(4G)$ fixes the lattice spacing at $\ell \approx 0.664\,\ell_P$, order-one in Planck units. The holographic entropy density $S/V \sim 1/m$ yields a cosmological constant estimate within a factor of 5.4 of the observed value, improving the standard quantum field theory prediction by 121 orders of magnitude. The discrete dimension converges to the continuum value with power-law corrections $\delta d \sim m^{-\beta}$ where $\beta \approx 1.5$–1.7, providing a quantitative discrete-continuum correspondence. All results follow from one combinatorial input with one free parameter.

INTRODUCTION

The Bekenstein-Hawking entropy $S = A/(4G\hbar)$ [1, 2] is one of the deepest results in theoretical physics: the information content of a black hole scales with its horizon *area*, not its volume. Deriving this area law from first principles has been a central goal of quantum gravity for five decades. Existing derivations require string theory [3], loop quantum gravity [4], entanglement entropy of quantum fields [5], or the AdS/CFT correspondence [6].

We present a derivation that requires none of these. The only input is a combinatorial fact about discrete partially ordered sets: the number of causally convex subsets of a d-dimensional lattice grows as $\exp(c_d \cdot m^{d-1})$, where m is the lattice extent and the exponent $d-1$ is determined by the spacetime dimension. Restricting to time-symmetric configurations reduces the exponent by exactly one, yielding area scaling. The Hawking temperature follows from a single derivative.

The approach is rooted in causal set theory [7, 8], which posits that spacetime is fundamentally a discrete partial order. The Noetherian ratio $\gamma = |\mathrm{CC}|/|\mathrm{Int}|$, introduced in the causal-algebraic geometry framework [17], serves as a combinatorial invariant that is strictly finer than all classical poset invariants. Prior approaches to entropy in causal sets have counted order intervals [9], maximal antichains [10], and causal links [11, 12]. These yield dimension estimators but not entropy laws: the number of intervals in $[m]^d$ is $|\mathrm{Int}| = (m(m+1)/2)^d = \Theta(m^{2d})$, giving $\log|\mathrm{Int}| = \Theta(\log m)$ — polynomial in m with no dimension-dependent exponential growth. Causally convex subsets are a richer combinatorial object — the convexity constraint encodes geometric information (the "shape" of a causal region) that simpler counts do not capture. The dimension law $\log|\mathrm{CC}| \sim m^{d-1}$ is specific to convex subsets and is the key to the area law derivation.

We emphasize at the outset the status of each result. The dimension law is a theorem for all $d \geq 2$, proved via a tiling inequality and formally verified [16]: the lower bound $\log|\mathrm{CC}([m]^d)| = \Omega(m^{d-1})$ follows from a super-multiplicativity argument across k^{d-1} tiles, and the upper bound $\log|\mathrm{CC}([m]^d)| = O(m^{d-1})$ from an ideal/filter injection. All physical results downstream — the area law, the Hawking temperature, the lattice spacing, and the cosmological constant estimate — inherit this rigorous status (modulo the physical interpretation of cylindrical subsets as equilibrium configurations, which is a kinematic identification, not a theorem).

THE DIMENSION LAW

Let $[m]^d = \{1, \ldots, m\}^d$ with the componentwise partial order: $(x_1, \ldots, x_d) \leq (y_1, \ldots, y_d)$ iff $x_i \leq y_i$ for all i. A subset $S \subseteq [m]^d$ is *causally convex* (also called order-convex in the combinatorics literature) if whenever $a, b \in S$ with $a \leq c \leq b$, then $c \in S$. Let $\mathrm{CC}([m]^d)$ denote the set of all causally convex subsets.

Dimension law (Theorem; proved for all $d \geq 2$).

$$\log|\mathrm{CC}([m]^d)| = \Theta(m^{d-1}). \tag{1}$$

The lower bound follows from a tiling inequality:

partitioning $[km]^d$ into k^{d-1} copies of $[m]^d$ gives $|\mathrm{CC}([km]^d)| \geq |\mathrm{CC}([m]^d)|^{\Theta(k^{d-1})}$. The upper bound follows from the ideal/filter injection $|\mathrm{CC}([m]^d)| \leq |\mathrm{downsets}([m]^d)|^2 \leq \binom{2m}{m}^{2(d-1)} = \exp(O(m^{d-1}))$. Both bounds are formally verified [16, 21]. Table I confirms convergence of the normalized ratio. For $d = 2$, the exact growth constant $\rho_2 = 16$ is proved; for $d \geq 3$, the $\Theta(m^{d-1})$ scaling is established but the exact constants c_d remain open.

The sequences $|\mathrm{CC}([m]^2)| = 2, 13, 114, 1146, 12578, \ldots$ and $|\mathrm{CC}([m]^3)| = 2, 101, 114797, 3071673482, \ldots$ have not been previously tabulated beyond the exact formulas of Barnette, Nichols, and Richmond [18] for small fixed column count. The asymptotic growth rates are, to the best of our knowledge, new.

The exponent $d-1$ means the number of bits needed to specify a convex configuration in d-dimensional discrete spacetime scales with the $(d-1)$-dimensional *boundary*, not the d-dimensional bulk. This is a combinatorial form of holography.

The dimension law has an independent structural precedent. Bollobás and Brightwell [19] proved that the height (longest chain) of a Poisson process in a box space of dimension d scales as $n^{1/d}$; Brightwell and Luczak [20] survey the mathematical theory. The height exponent $1/d$ and our convex-subset exponent $d-1$ are different invariants detecting the same underlying dimension, providing evidence that dimensional detection from poset combinatorics is robust.

TABLE I. Computed values of $|\mathrm{CC}([m]^d)|$ and the normalized growth ratio $\log_2 |\mathrm{CC}|/m^{d-1}$. The $d = 2$ values for $m \leq 4$ are kernel-verified; the $d = 3$ and $d = 4$ values are obtained by independent C and Python enumerations.

d	m	$\lvert\mathrm{CC}([m]^d)\rvert$	$\log_2 /m^{d-1}$
2	1	2	1.00
2	2	13	1.85
2	3	114	2.28
2	4	1,146	2.54
2	∞	—	4.00[a]
3	1	2	1.00
3	2	101	1.66
3	3	114,797	1.87
3	4	3,071,673,482	1.97
4	1	2	1.00
4	2	3,938	1.49

[a] Upper bound $\rho_2 \leq 16$ fully verified; lower bound verified except for one auxiliary lemma. See [16].

THE AREA LAW

A *cylindrical* convex subset of $[m]^d$ is one of the form $C \times I$, where $C \subseteq [m]^{d-1}$ is a causally convex subset of the spatial lattice and $I = [t_1, t_2] \subseteq [m]$ is a contiguous time interval. The count factorizes:

$$|\mathrm{Cyl}([m]^d)| = |\mathrm{CC}([m]^{d-1})| \times \binom{m+1}{2}. \tag{2}$$

Applying the dimension law (1) to the spatial factor:

$$\log |\mathrm{Cyl}([m]^d)| = c_{d-1} \cdot m^{d-2} + O(\log m). \tag{3}$$

For $d = 4$, Eq. (3) gives $\log |\mathrm{Cyl}([m]^4)| \sim c_3 \cdot m^2$, where m^2 is the spatial area. Table II confirms convergence.

Physical interpretation. We identify cylindrical convex subsets with equilibrium black hole configurations. A Schwarzschild black hole in thermal equilibrium has a time-independent metric: its spatial interior (the region causally disconnected from future infinity) is constant on each Cauchy slice. In the discrete setting, a time-independent spatial configuration is precisely a cylindrical subset $C \times [t_1, t_2]$: the cross-section C is the same at every time step. Causal convexity of C ensures that the interior is causally consistent. The time interval represents the equilibrium duration. This identification is kinematic, not dynamical.

The full convex entropy m^{d-1} includes temporally varying (non-equilibrium) configurations; restricting to equilibrium removes exactly one power of m. The area law is specific to the cylindrical class: diamond-shaped subsets $[p, q]$ yield entropy linear in m regardless of d, because a diamond is specified by two points (a one-dimensional choice). Time-symmetry is the essential ingredient producing area scaling.

DIMENSION FROM THE AREA LAW

The dimension d is not an input to the area law — it is a consequence. The width scaling law [17] gives, for any poset P on n elements with width w:

$$\log |\mathrm{CC}(P)| \;\leq\; w \cdot \log(\lceil n/w \rceil^2 + 1). \tag{4}$$

TABLE II. Cylindrical convex subsets of $[m]^4$ and the area-law ratio.

m	$\log_2 \lvert\mathrm{Cyl}([m]^4)\rvert$	$\log_2 /m^2$
1	1.00	1.00
2	8.23	2.06
3	19.39	2.15
4	34.84	2.18

For the grid $[m]^d$ with $n = m^d$, the width is $w = m^{d-1} = n^{(d-1)/d}$, and the dimension law gives $\log|\mathrm{CC}| = \Theta(m^{d-1}) = \Theta(n^{(d-1)/d})$. Conversely, if any discrete spacetime on n elements has $\log|\mathrm{CC}| = \Theta(n^\alpha)$, then by Eq. (4) the width must satisfy $w = \Omega(n^\alpha/\log n)$, forcing $\alpha \le (d-1)/d$ for any poset embeddable in d dimensions. The exponent α determines the dimension:

$$d = \frac{1}{1-\alpha}. \tag{5}$$

The Bekenstein-Hawking area law states $S \sim A \sim m^{d-2}$ for cylindrical configurations. Since the cylindrical restriction reduces the exponent by one, this corresponds to $\alpha = (d-1)/d$ with $d-1 = 3$, hence $d = 4$. No other dimension produces area scaling of equilibrium entropy in a discrete spacetime whose full convex entropy obeys the dimension law.

This is a uniqueness result: a discrete spacetime whose convex-subset entropy obeys the Bekenstein-Hawking area law *must* have width scaling as $n^{3/4}$, which is the hallmark of 4-dimensional structure. The dimension is not chosen — it is forced by the entropy scaling.

ENTROPY HIERARCHY

Three natural subclasses yield three distinct scalings (all verified):

TABLE III. Entropy hierarchy for d-dimensional discrete spacetime.

Subset class	Scaling	Physical meaning
Diamonds $[p,q]$	m	Observer horizon
Cylindrical $C \times I$	m^{d-2}	Equilibrium (area law)
All convex	m^{d-1}	Full dynamics

Each class differs from the next by one power of m. This hierarchy is reminiscent of holographic renormalization, where each step along the holographic direction adds degrees of freedom.

HAWKING TEMPERATURE

The equilibrium entropy $S(m) = c_3 m^2$ for $d = 4$ gives:

$$T = \left(\frac{\partial S}{\partial m}\right)^{-1} = \frac{1}{2c_3\, m}. \tag{6}$$

Since $M \sim r_s \sim m\ell$, this gives $T \sim 1/M$: the Hawking scaling [2]. The finite-difference T^{-1}/m converges as $2.50, 2.58, 2.68 \to 2c_3 \approx 2.77$, confirming $T \sim 1/m$ from the data. Convergence corrections are $O(m^{-\beta})$ with $\beta \approx 1.5$ (Sec.).

LATTICE SPACING

Matching $S_{\mathrm{comb}} = c_3 r_s^2/\ell^2$ to $S_{\mathrm{BH}} = \pi r_s^2/\ell_P^2$:

$$\ell = \ell_P\sqrt{\frac{c_3}{\pi}} \approx 0.664\,\ell_P. \tag{7}$$

One free parameter, fixed once. The temperature is then a prediction. For a solar-mass black hole, $S_{\mathrm{comb}}/S_{\mathrm{BH}} = 1.089$ at $m = 4$, converging to 1.000.

COSMOLOGICAL CONSTANT

The entropy density $s = S/V = c_3/m$ (verified: $s \cdot m = 0.69, 1.43, 1.49, 1.51 \to 1.386$). The vacuum energy $\rho \sim Ts \sim 1/m^2$. With $m = L/\ell$, $L = H^{-1}$:

$$\rho_{\mathrm{vac}} \sim H^2 M_P^2. \tag{8}$$

This is the holographic dark energy scaling [13]. Those authors *assumed* the area bound; here it is *derived*. The application to the cosmological horizon is justified by the Gibbons-Hawking result that de Sitter horizons carry area-proportional entropy [14].

TABLE IV. Cosmological constant: three predictions.

Prediction	ρ (kg/m^3)	Discrepancy
QFT	5.2×10^{96}	10^{122}
This work	3.4×10^{-26}	5.4×
Observed	6.3×10^{-27}	—

The remaining factor 5.4 may reflect finite-m corrections to the dimension law. The discrete-continuum correspondence (Sec.) shows that the effective dimension converges as $\delta d \sim m^{-\beta}$ with $\beta \approx 1.5$–1.7; the same power-law corrections propagate through the entropy density to the vacuum energy estimate, suggesting that the factor 5.4 is $O(m^{-\beta})$ and vanishes in the continuum limit. More concretely, the exact growth constant c_3 for the $d = 3$ grid (analogous to $\rho_2 = 16$ for $d = 2$) would sharpen the entropy coefficient and hence the Λ prediction; determining c_3 is the single most impactful open problem for this estimate. We do not claim exactness; the significance is that holographic scaling emerges from the dimension law, reducing 10^{122} to 5.4.

DISCRETE-CONTINUUM CORRESPONDENCE

The effective dimension $d_{\mathrm{eff}}(m) = 1 + \Delta\log|\mathrm{CC}|/\Delta\log m$ converges to d from above:

The best-fit exponent is $\beta \approx 1.5$–1.65, between boundary corrections ($\beta = 1$) and the heat kernel ($\beta = 2$) [15]. Achieving 1% dimension accuracy requires $m \ge 12$ (∼1,700 elements for $d = 3$).

TABLE V. Dimensional flow for $d = 3$: $\delta d = d_{\text{eff}} - d$.

Transition	d_{eff}	δd	$\delta d \cdot m^{1.5}$
$1 \to 2$	3.74	0.74	2.09
$2 \to 3$	3.28	0.28	1.45
$3 \to 4$	3.19	0.19	1.52

BOUNDARY FREEDOM

We tested whether the convex subset entropy density $\ln|\text{CC}|/n$ detects spatial geometry using matched-bottleneck grids (same minimum width 6, same number of rows 10; element counts differ because width profiles differ):

TABLE VI. Curvature test: matched bottleneck.

Shape	n	$\lvert\text{CC}\rvert$	$\ln\lvert\text{CC}\rvert/n$
Saddle [10,9,...,6,6,...,9,10]	80	597,739,578	0.253
Flat [6×10]	60	9,358,669	0.268
Sphere [6,7,...,10,10,...,7,6]	80	55,019,893	0.223

The saddle ($|\text{CC}|_{\text{saddle}} \approx 11 \times |\text{CC}|_{\text{sphere}}$) concentrates width at the temporal boundary, where nesting constraints are weakest; the sphere buries width in the constrained interior. The entropy density detects the distribution of degrees of freedom relative to the causal boundary — a holographic property consistent with the area law — rather than local Ricci curvature directly.

DISCUSSION

TABLE VII. Summary: proved versus derived.

Result	Status
Dimension law, all $d \geq 2$	Proved
Area law implies $d = 4$	Corollary
Hawking $T \sim 1/m$	Derivative
$\ell \approx 0.664\,\ell_P$	Matching
Λ within 5.4×	Thermodynamic identity
$\beta \approx 1.5$	Computed
Entropy density detects boundary	Computed
Exact constants c_d, $d \geq 3$	Open

The derivation chain — dimension law → cylindrical restriction → area law → temperature → lattice spacing → entropy density → cosmological constant — uses one combinatorial input and one free parameter. The dimension law, which was the sole conjectural link in an earlier version of this work, is now a theorem for all $d \geq 2$. The converse direction (Sec.) shows that the area law forces $d = 4$: the derivation is not only sufficient but unique. No quantum field theory, no string theory, no Bogoliubov transformations.

Open questions. (i) Extension to Poisson sprinklings. (ii) Local curvature detection via neighborhood-restricted γ. (iii) Convergence exponent on periodic lattices. (iv) Exact constants c_d for $d \geq 3$. (v) Reducing the factor 5.4 in Λ.

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